

IMPERIAL

DEPARTMENT OF CIVIL AND ENVIRONMENTAL ENGINEERING

CIVE70100 RESEARCH PROJECT – STRUCTURES 2025–2026

Developing design formulae for flexural buckling of stainless steel columns using machine learning techniques

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Afshan and Gardner (2013) reported experimental results for cold-formed ferritic stainless steel hollow sections, building on earlier column and beam tests (Gardner and Nethercot, 2004).

2 Literature review

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Bibliography

- Afshan, S. and Gardner, L. (May 2013). Experimental Study of Cold-Formed Ferritic Stainless Steel Hollow Sections. In: *Journal of Structural Engineering* 139.5, pp. 717–728. ISSN: 0733-9445, 1943-541X. DOI: [10.1061/\(ASCE\)ST.1943-541X.0000580](https://doi.org/10.1061/(ASCE)ST.1943-541X.0000580). Available at: <https://ascelibrary.org/doi/10.1061/%28ASCE%29ST.1943-541X.0000580> (Accessed: 06/13/2026).
- Gardner, L. and Nethercot, D. (Sept. 2004). Experiments on Stainless Steel Hollow Sections—Part 2: Member Behaviour of Columns and Beams. In: *Journal of Constructional Steel Research* 60.9, pp. 1319–1332. ISSN: 0143974X. DOI: [10.1016/j.jcsr.2003.11.007](https://doi.org/10.1016/j.jcsr.2003.11.007). Available at: <https://linkinghub.elsevier.com/retrieve/pii/S0143974X03002165> (Accessed: 06/18/2026).